Web Dev Mastery

🎬 Bootstrap 5 Mastery Course – With 5 Real-World Projects

✅ **Prerequisites:** Basic HTML & CSS knowledge

📚 What We Cover

1. What Bootstrap is and why we use it
2. How to set up Bootstrap in your projects (CDN & Local installation)
3. Understanding Bootstrap Typography, Colors, and Utilities
4. Bootstrap Components (Buttons, Badges, Alerts, Cards, Navbar, etc.)
5. Bootstrap Forms and Form Validation
6. Bootstrap Grid System & Responsive Layouts
7. Bootstrap Helpers and Utility Classes (Spacing, Borders, Shadows, etc.)
8. Bootstrap Modals, Collapse, Carousel, and Other Interactive Components

🧠 PART 1: Introduction to Bootstrap

- What is Bootstrap?
- Why Bootstrap is popular (fast development, responsive design, prebuilt components)
- Real-world usage examples (Airbnb, Spotify, etc.)

🎥 **Demo:** Exploring Bootstrap's official documentation

🧠 PART 2: Setting up Bootstrap

- Installing via **CDN**
- Installing locally
- Integrating with HTML files
- Folder structure for Bootstrap projects

🎥 **Demo:** Creating your first HTML page with Bootstrap

🧠 PART 3: Typography, Colors & Utilities

- Bootstrap typography classes (h1–h6, display headings, lead text)
- Text alignment, transformation, and weight
- Color utilities (text & background colors)
- Spacing utilities (margin, padding)

🎥 **Demo:** Styled text section for a portfolio page

🧠 PART 4: Bootstrap Components

- Buttons (default, outline, sizes)
- Badges & Alerts
- Cards – layouts and images
- List groups

🎥 **Demo:** Creating a blog post card layout

🧠 PART 5: Bootstrap Forms

- Input fields, checkboxes, radio buttons
- Input groups and addons
- Form validation styles
- Select menus and custom controls

🎥 **Demo:** Sign-up form for an event page

🧠 PART 6: Bootstrap Grid System & Responsive Layout

- Containers, rows, columns
- Breakpoints (`sm`, `md`, `lg`, `xl`, `xxl`)
- Nesting columns
- Auto-layout columns
- Responsive design principles

🎥 **Demo:** Creating a responsive 3-column services section

🧠 PART 7: Bootstrap Components (Part 2)

- Navbar with dropdowns
- Breadcrumbs & Pagination
- Modals
- Accordion & Collapse
- Carousel

🎥 **Demo:** Interactive image carousel for a travel site

🧠 PART 8: Bootstrap Helpers & Utility Classes

- Border utilities
- Shadows
- Position utilities
- Flex utilities
- Display & Visibility classes

🎥 **Demo:** Sticky navbar with shadow

🧠 PART 1: Introduction to Bootstrap

What is Bootstrap?

A free, open-source **front-end framework** with ready-made **HTML, CSS & JS components** to quickly build **responsive, mobile-first websites**.

Why it's Popular

- **Fast Development** – Prebuilt styles & UI elements save time.
- **Responsive** – Grid system & mobile-first design look great on all devices.
- **Prebuilt Components** – Buttons, navbars, modals, forms, carousels, etc.
- **Cross-browser Support** – Works across major browsers.
- **Huge Community** – Tons of templates, themes, and support.

Real-world Usage

Used by **Airbnb, Spotify, NASA, Vogue**, and many startups for quick, professional UI.

🎥 Demo Idea

Visit getbootstrap.com →

- Check *Getting Started* for the CDN link.
- Explore *Components* like Navbar, Cards, Buttons.
- Try *Grid System* to make layouts responsive in minutes.

🧠 PART 2 — Setting up Bootstrap

Below is a clear, practical guide you can use in class or as a cheat-sheet.

Installing via CDN

Fastest way — great for demos / prototypes.

Steps

1. Copy the CSS `<link>` into `<head>`.
2. Copy the JS bundle before `</body>` (bundle includes Popper for dropdowns/tooltips).
3. (Optional) Grab the exact CDN links from **getbootstrap.com** to get the latest stable version and SRI hashes.

Example

```
<!-- head -->

<link
href="https://cdn.jsdelivr.net/npm/bootstrap@5/dist/css/bootstrap.min.css
" rel="stylesheet">



<!-- before </body> -->

<script
src="https://cdn.jsdelivr.net/npm/bootstrap@5/dist/js/bootstrap.bundle.mi
n.js"></script>
```

Why CDN?

- Instant, no build step.
- Good for learning, quick pages, and prototypes.

Installing locally (npm / yarn)

Better for production, customization, or using SCSS and a bundler (Vite/webpack).

Commands

```
npm init -y

npm install bootstrap

# (optional) install sass if you want to compile SCSS

npm install --save-dev sass
```

Use with a bundler (simple example)

```
src/index.js

import 'bootstrap/dist/css/bootstrap.min.css';

import 'bootstrap/dist/js/bootstrap.bundle.min.js';
```

Then run your bundler (Vite/webpack/parcel) — it will include Bootstrap in your build.

Using SCSS for customization

```
// src/styles.scss

@import "bootstrap/scss/bootstrap";

// override variables before import if needed:
```

```
$primary: #1a73e8;

@import "bootstrap/scss/bootstrap";
```

Build with `npx sass src/styles.scss dist/css/styles.css`.

Why local?

- You can customize variables, tree-shake, and ship CSS/JS bundled with your app.

Integrating with HTML files

Basic rules

- Put responsive meta tag in `<head>`: `<meta name="viewport" content="width=device-width, initial-scale=1">`
- Load CSS in `<head>`.
- Load `bootstrap.bundle.min.js` before `</body>` so interactive components work.
- No jQuery needed for Bootstrap 5+ (use bundle which includes Popper).

Small static site example (local files copied)

```
/project
  /css
    bootstrap.min.css
    styles.css
  /js
    bootstrap.bundle.min.js
```

```
    main.js
  index.html
```

In `index.html` link `css/bootstrap.min.css` and `js/bootstrap.bundle.min.js`.

Folder structure (recommended)

Tiny/simple project

```
/project
  index.html
  /css
    styles.css
  /js
    main.js
  /assets
    /img
```

Modern app (with bundler)

```
/project
  /src
    index.html
    main.js         <-- imports bootstrap CSS/JS
    /scss
```

```
      styles.scss   <-- imports bootstrap SCSS
    /components
  /public
  package.json
  vite.config.js (or webpack)
  /dist (build output)
```

Large / production

- Keep third-party libs in `node_modules` and bundle/minify into `/dist`.
- Keep custom theme variables in a `_variables.scss` and import before Bootstrap.

🎥 Demo — Create your first HTML page with Bootstrap (CDN)

Copy this into `index.html` and open in a browser:

```
<!doctype html>
<html lang="en">
<head>
  <meta charset="utf-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1" />
  <title>Bootstrap Starter</title>
```

```
  <!-- Bootstrap CSS (CDN) -->

  <link
href="https://cdn.jsdelivr.net/npm/bootstrap@5/dist/css/bootstrap.min.css
" rel="stylesheet" />

</head>

<body>



<nav class="navbar navbar-expand-lg navbar-light bg-light">

  <div class="container">

    <a class="navbar-brand" href="#">MySite</a>

    <button class="navbar-toggler" type="button"
data-bs-toggle="collapse" data-bs-target="#nav">

      <span class="navbar-toggler-icon"></span>

    </button>

    <div class="collapse navbar-collapse" id="nav">

      <ul class="navbar-nav ms-auto">

        <li class="nav-item"><a class="nav-link" href="#">Home</a></li>

        <li class="nav-item"><a class="nav-link" href="#">Docs</a></li>

      </ul>

    </div>

  </div>

</nav>



<main class="container py-5">
```

```
  <div class="row g-4">
    <div class="col-md-8">
      <div class="p-4 border rounded">
        <h1>Welcome to Bootstrap</h1>
        <p class="lead">Responsive layout and components ready to
use.</p>
        <button class="btn btn-primary">Get started</button>
      </div>
    </div>
    <div class="col-md-4">
      <div class="p-3 border rounded">Sidebar / card</div>
    </div>
  </div>
</main>


<!-- Bootstrap JS Bundle (includes Popper) -->
<script
src="https://cdn.jsdelivr.net/npm/bootstrap@5/dist/js/bootstrap.bundle.mi
n.js"></script>
</body>
</html>
```

Quick checklist

- ✅ Add viewport meta tag.
- ✅ Use CDN for quick demos.
- ✅ Use npm + bundler for real apps and SCSS customization.
- ✅ Include `bootstrap.bundle.min.js` for components that need JS.
- ✅ Copy official CDN links from **getbootstrap.com** when producing shared code.

🧠 PART 3: Typography, Colors & Utilities

1️⃣ Bootstrap Typography Classes

Bootstrap gives you **ready-to-use text styles** without writing custom CSS.

Headings

- Normal HTML headings: `<h1>` to `<h6>` already styled by Bootstrap.
- Class-based headings: You can apply heading styles to other elements:

```
<p class="h1">Heading styled like h1</p>
```

Display Headings (bigger & bolder than normal headings)

```
<h1 class="display-1">Display 1</h1>

<h1 class="display-2">Display 2</h1>
```

```
...
<h1 class="display-6">Display 6</h1>
```

- Used for **hero sections** or **big titles**.

Lead Text

- For slightly larger, lighter text for introductions.

```
<p class="lead">This is lead text for emphasis.</p>
```

2️⃣ Text Alignment, Transformation & Weight

Text Alignment

- `text-start` → Align left
- `text-center` → Align center
- `text-end` → Align right

```
<p class="text-center">Centered text</p>
```

Text Transformation

- `text-lowercase` → all lowercase
- `text-uppercase` → ALL UPPERCASE
- `text-capitalize` → Capitalize Each Word

```
<p class="text-uppercase">hello world</p>
```

Font Weight & Style

- `fw-bold` → Bold
- `fw-semibold` → Semi-bold
- `fw-normal` → Normal
- `fw-light` → Light
- `fst-italic` → Italic
- `fst-normal` → Remove italic

```
<p class="fw-bold">Bold text</p>

<p class="fst-italic">Italic text</p>
```

3️⃣ Color Utilities

Bootstrap has **text color** and **background color** classes based on its theme colors.

Text Colors

```
<p class="text-primary">Primary text</p>

<p class="text-success">Success text</p>

<p class="text-danger">Danger text</p>

<p class="text-warning">Warning text</p>

<p class="text-info">Info text</p>
```

```
<p class="text-muted">Muted gray text</p>

<p class="text-white bg-dark">White text on dark background</p>
```

Background Colors

```
<div class="bg-primary text-white p-3">Primary background</div>

<div class="bg-success text-white p-3">Success background</div>

<div class="bg-warning text-dark p-3">Warning background</div>
```

4️⃣ Spacing Utilities (Margin & Padding)

This is where things like `my-4` come in.

Format:

```
m | p  (type: margin or padding)

t | b | s | e | x | y | (side: top, bottom, start, end, horizontal,
vertical)

0–5 (size: 0 smallest, 5 largest)
```

Type

- `m` → margin
- `p` → padding

Side

- `t` → top
- `b` → bottom
- `s` → start (left in LTR layout)
- `e` → end (right in LTR layout)
- `x` → left + right
- `y` → top + bottom
- *(no letter)* → all sides

Size

- `0` → `0` spacing
- `1` → `0.25rem` (~4px)
- `2` → `0.5rem` (~8px)
- `3` → `1rem` (~16px)
- `4` → `1.5rem` (~24px)
- `5` → `3rem` (~48px)

Examples

```
<div class="mt-3">Margin Top 1rem</div>

<div class="mb-5">Margin Bottom 3rem</div>

<div class="px-4">Padding Left+Right 1.5rem</div>
```

```
<div class="py-2">Padding Top+Bottom 0.5rem</div>

<div class="m-0">No margin</div>
```

✅ `my-4` = **Margin Y-axis (top + bottom) 1.5rem (~24px)**.

✅ `px-3` = **Padding X-axis (left + right) 1rem (~16px)**.

✅ `p-2` = Padding all sides **0.5rem (~8px)**.

🎥 Demo — Styled Text Section for a Portfolio Page

```
<section class="container my-5">

  <h1 class="display-4 fw-bold text-center text-primary">Suman Malakar</h1>

  <p class="lead text-center text-muted mb-4">MERN Stack Developer & Technical Trainer</p>

  <p class="text-center px-3">

    Passionate about building modern, responsive web applications with clean UI and efficient backend systems.

  </p>

  <div class="text-center mt-4">

    <a href="#" class="btn btn-success px-4 py-2">View Projects</a>

  </div>

</section>
```

What's happening here:

- `display-4` → large heading
- `fw-bold` → bold
- `my-5` → vertical margin 3rem
- `mb-4` → bottom margin 1.5rem
- `px-3` → padding left/right 1rem on paragraph
- `mt-4` → top margin 1.5rem for button
- Colors applied using `text-primary`, `text-muted`, and `btn-success`.

🧠 PART 4: Bootstrap Components

1️⃣ Buttons

Bootstrap buttons use `.btn` + a **color class**.

Default Buttons

```
<button class="btn btn-primary">Primary</button>

<button class="btn btn-success">Success</button>

<button class="btn btn-danger">Danger</button>

<button class="btn btn-warning">Warning</button>

<button class="btn btn-info">Info</button>

<button class="btn btn-dark">Dark</button>

<button class="btn btn-secondary">Secondary</button>
```

```
<button class="btn btn-light">Light</button>
```

Outline Buttons

```
<button class="btn btn-outline-primary">Outline Primary</button>

<button class="btn btn-outline-success">Outline Success</button>
```

These have colored borders and text, but transparent background.

Sizes

```
<button class="btn btn-primary btn-lg">Large</button>

<button class="btn btn-primary btn-sm">Small</button>
```

2️⃣ Badges & Alerts

Badges (small highlighted text)

```
<h1>Notifications <span class="badge bg-danger">5</span></h1>

<button class="btn btn-primary">

  Messages <span class="badge bg-light text-dark">4</span>

</button>
```

Alerts (highlighted message boxes)

```
<div class="alert alert-success" role="alert">

  This is a success alert!
```

```
</div>
<div class="alert alert-warning" role="alert">
  Warning! Please check your input.
</div>
```

- You can also add a close button with `.alert-dismissible`.

3️⃣ Cards

Cards are flexible content containers with an optional header, footer, image, and body.

Basic Card

```
<div class="card" style="width: 18rem;">
  <img src="img.jpg" class="card-img-top" alt="...">
  <div class="card-body">
    <h5 class="card-title">Card title</h5>
    <p class="card-text">Some description here...</p>
    <a href="#" class="btn btn-primary">Read more</a>
  </div>
</div>
```

Card Layouts

- Use **Grid system** (`row` + `col`) to make multiple cards side by side.
- Add `card-group` or `d-flex gap-3` for equal height.

4️⃣ List Groups

Simple lists with Bootstrap styling.

```
<ul class="list-group">
  <li class="list-group-item">First item</li>
  <li class="list-group-item">Second item</li>
  <li class="list-group-item active">Active item</li>
  <li class="list-group-item disabled">Disabled item</li>
</ul>
```

You can also link them:

```
<div class="list-group">
  <a href="#" class="list-group-item list-group-item-action">Link 1</a>
  <a href="#" class="list-group-item list-group-item-action active">Link 2</a>
</div>
```

🎥 Demo — Blog Post Card Layout

```
<div class="container my-5">
  <div class="row g-4">
    <div class="col-md-4">
      <div class="card h-100">
        <img src="blog1.jpg" class="card-img-top" alt="Blog 1">
        <div class="card-body">
          <h5 class="card-title">Blog Title 1</h5>
          <p class="card-text">Quick summary of the blog post goes
here...</p>
          <a href="#" class="btn btn-primary btn-sm">Read More</a>
        </div>
        <div class="card-footer text-muted">
          <span class="badge bg-info text-dark">Tech</span>
          <span class="badge bg-success">5 min read</span>
        </div>
      </div>
    </div>
    <!-- Repeat for more cards -->
  </div>
</div>
```

Key Features Here:

- `.row` + `.col-md-4` → Responsive layout (3 cards in a row on desktop).
- `.card h-100` → Makes all cards equal height.
- `.badge` in footer → Category & read time.
- `.g-4` → Gap between columns.

🧠 PART 5: Bootstrap Forms

1️⃣ Input Fields, Checkboxes, Radio Buttons

Basic Input Field

```
<form>
  <div class="mb-3">
    <label for="name" class="form-label">Full Name</label>
    <input type="text" class="form-control" id="name" placeholder="Enter your name">
  </div>
</form>
```

- `.form-label` → Styling for labels
- `.form-control` → Makes input full-width with padding & border styling

Checkbox

```
<div class="form-check">
  <input class="form-check-input" type="checkbox" id="subscribe">
  <label class="form-check-label" for="subscribe">Subscribe to newsletter</label>
</div>
```

- `.form-check` → Wrapper for checkbox or radio
- `.form-check-input` → Styles the checkbox itself
- `.form-check-label` → Styles the label

Radio Button

```
<div class="form-check">
  <input class="form-check-input" type="radio" name="gender" id="male" checked>
  <label class="form-check-label" for="male">Male</label>
</div>
<div class="form-check">
  <input class="form-check-input" type="radio" name="gender" id="female">
```

```
  <label class="form-check-label" for="female">Female</label>
</div>
```

- Use **same `name`** to group radios so only one can be selected.

2️⃣ Input Groups & Addons

Used when you want to **add text, icons, or buttons** inside the input.

```
<div class="input-group mb-3">
  <span class="input-group-text">@</span>
  <input type="text" class="form-control" placeholder="Username">
</div>
```

```
<div class="input-group">
  <input type="text" class="form-control" placeholder="Recipient's
username">
  <button class="btn btn-outline-secondary" type="button">Send</button>
</div>
```

- `.input-group-text` → Static text or icon before/after input
- Can also add **buttons** to the group.

3️⃣ Form Validation Styles

Bootstrap uses **classes** to indicate valid or invalid inputs.

```
<form class="needs-validation" novalidate>

  <div class="mb-3">

    <label for="email" class="form-label">Email address</label>

    <input type="email" class="form-control is-valid" id="email"
required>

    <div class="valid-feedback">Looks good!</div>

  </div>

  <div class="mb-3">

    <input type="text" class="form-control is-invalid"
placeholder="Invalid example">

    <div class="invalid-feedback">Please enter a valid value.</div>

  </div>

</form>
```

- `is-valid` → Green border & success message
- `is-invalid` → Red border & error message
- `.valid-feedback` / `.invalid-feedback` → Messages

4️⃣ Select Menus & Custom Controls

```
<select class="form-select" aria-label="Default select example">
  <option selected>Choose an option</option>
  <option value="1">Option One</option>
  <option value="2">Option Two</option>
</select>
```

- `.form-select` → Bootstrap-styled dropdowns.

🎥 Demo — Sign-up Form for an Event Page

```
<div class="container my-5">
  <h2 class="mb-4">Event Sign-Up</h2>
  <form class="needs-validation" novalidate>

    <!-- Name -->
    <div class="mb-3">
      <label for="fullname" class="form-label">Full Name</label>
      <input type="text" class="form-control" id="fullname"
placeholder="John Doe" required>
      <div class="invalid-feedback">Please enter your full name.</div>
    </div>
```

```
    <!-- Email -->
    <div class="mb-3">
      <label for="email" class="form-label">Email address</label>
      <input type="email" class="form-control" id="email"
placeholder="name@example.com" required>
      <div class="invalid-feedback">Enter a valid email address.</div>
    </div>


    <!-- Gender -->
    <div class="mb-3">
      <label class="form-label">Gender</label>
      <div class="form-check">
        <input class="form-check-input" type="radio" name="gender"
id="male">
        <label class="form-check-label" for="male">Male</label>
      </div>
      <div class="form-check">
        <input class="form-check-input" type="radio" name="gender"
id="female">
        <label class="form-check-label" for="female">Female</label>
      </div>
    </div>


    <!-- Ticket Type -->
```

```
    <div class="mb-3">
      <label for="ticket" class="form-label">Ticket Type</label>
      <select class="form-select" id="ticket" required>
        <option value="">Choose...</option>
        <option value="standard">Standard</option>
        <option value="vip">VIP</option>
      </select>
      <div class="invalid-feedback">Please select a ticket type.</div>
    </div>

    <!-- Newsletter -->
    <div class="form-check mb-3">
      <input class="form-check-input" type="checkbox" id="subscribe">
      <label class="form-check-label" for="subscribe">
        Subscribe to newsletter
      </label>
    </div>

    <!-- Submit -->
    <button class="btn btn-primary" type="submit">Sign Up</button>
  </form>
</div>
```

Key Bootstrap Classes Used Here:

- `.form-control` → Inputs & text areas
- `.form-check` → Checkbox & radio wrapper
- `.form-select` → Styled dropdown
- `.is-valid` / `.is-invalid` → Validation styles
- `.input-group` → Add icons/text to inputs
- `.mb-3` → Margin bottom for spacing

🧠 PART 6: Bootstrap Grid System & Responsive Layout

1️⃣ Containers, Rows, Columns

Bootstrap's **Grid System** is based on **12 columns** per row.
Everything starts inside a **container** → rows → columns.

Containers

- `.container` → Fixed-width, changes at breakpoints
- `.container-fluid` → Full-width, always spans 100%

```
<div class="container">
  <div class="row">
    <div class="col">Column 1</div>
    <div class="col">Column 2</div>
```

```
    <div class="col">Column 3</div>
  </div>
</div>
```

2 Breakpoints (Responsive Sizes)

Bootstrap uses **5 main breakpoints**:

Abbreviation	Min Width	Example class
`sm`	≥576px	`.col-sm-6`
`md`	≥768px	`.col-md-4`
`lg`	≥992px	`.col-lg-3`
`xl`	≥1200px	`.col-xl-2`
`xxl`	≥1400px	`.col-xxl-1`

Example:

```
<div class="col-12 col-md-6 col-lg-4">
  Responsive column
</div>
```

- On mobile → `col-12` (full width)
- On tablet → `col-md-6` (half width)
- On desktop → `col-lg-4` (one-third width)

3️⃣ Nesting Columns

You can place **rows inside columns** for more complex layouts.

```
<div class="row">
  <div class="col-md-6">
    <div class="row">
      <div class="col-6">Nested 1</div>
      <div class="col-6">Nested 2</div>
    </div>
  </div>
  <div class="col-md-6">Column 2</div>
</div>
```

4️⃣ Auto-layout Columns

If you don't set a number, columns share space equally.

```
<div class="row">
  <div class="col">Equal 1</div>
```

```
  <div class="col">Equal 2</div>
  <div class="col">Equal 3</div>
</div>
```

Or mix with fixed columns:

```
<div class="row">
  <div class="col-6">Half width</div>
  <div class="col">Auto fill</div>
</div>
```

5️⃣ Responsive Design Principles

- **Mobile-first** → Start with `.col-12` and scale up
- Use breakpoints to adjust layout for tablets, desktops, large screens
- Keep content readable (avoid too many small columns on mobile)
- Test by resizing browser or using Chrome DevTools

🎥 Demo — Responsive 3-column Services Section

```
<div class="container my-5">
  <h2 class="text-center mb-4">Our Services</h2>
  <div class="row g-4">
```

```
    <div class="col-12 col-md-6 col-lg-4">
      <div class="p-4 border rounded text-center">
        <h4>Web Development</h4>
        <p>We build responsive, modern websites tailored to your
needs.</p>
      </div>
    </div>

    <div class="col-12 col-md-6 col-lg-4">
      <div class="p-4 border rounded text-center">
        <h4>UI/UX Design</h4>
        <p>Beautiful and user-friendly designs for your brand.</p>
      </div>
    </div>

    <div class="col-12 col-md-6 col-lg-4">
      <div class="p-4 border rounded text-center">
        <h4>SEO Optimization</h4>
        <p>Improve your website’s ranking and visibility in search
engines.</p>
      </div>
    </div>
```

```
  </div>

</div>
```

What Happens Here:

- On **mobile** → 1 column (`col-12`)
- On **tablet** → 2 columns (`col-md-6`)
- On **desktop** → 3 columns (`col-lg-4`)
- `.g-4` → gap between items
- `.p-4` → padding inside boxes

🧠 PART 7: Bootstrap Components

1️⃣ Navbar with Dropdowns

The **navbar** is a responsive navigation header with links, logos, and menus.

```
<nav class="navbar navbar-expand-lg navbar-dark bg-dark">

  <div class="container-fluid">

    <a class="navbar-brand" href="#">TravelSite</a>

    <button class="navbar-toggler" type="button"
data-bs-toggle="collapse" data-bs-target="#mainNav">

      <span class="navbar-toggler-icon"></span>

    </button>
```

```
    <div class="collapse navbar-collapse" id="mainNav">

      <ul class="navbar-nav ms-auto">

        <li class="nav-item"><a class="nav-link" href="#">Home</a></li>

        <li class="nav-item"><a class="nav-link"
href="#">Destinations</a></li>



        <!-- Dropdown -->

        <li class="nav-item dropdown">

          <a class="nav-link dropdown-toggle" href="#"
data-bs-toggle="dropdown">Tours</a>

          <ul class="dropdown-menu">

            <li><a class="dropdown-item" href="#">Adventure</a></li>

            <li><a class="dropdown-item" href="#">Luxury</a></li>

          </ul>

        </li>



        <li class="nav-item"><a class="nav-link"
href="#">Contact</a></li>

      </ul>

    </div>

  </div>

</nav>
```

- `.navbar-expand-lg` → Expands on large screens, collapses on mobile
- `.navbar-dark bg-dark` → Dark style navbar
- `.dropdown` → Parent menu item with `.dropdown-menu`

2️⃣ Breadcrumbs & Pagination

Breadcrumbs

```
<nav aria-label="breadcrumb">
  <ol class="breadcrumb">
    <li class="breadcrumb-item"><a href="#">Home</a></li>
    <li class="breadcrumb-item active" aria-current="page">Europe Tours</li>
  </ol>
</nav>
```

- Shows **navigation path** to current page

Pagination

```
<nav>
  <ul class="pagination justify-content-center">
    <li class="page-item disabled"><a class="page-link">Previous</a></li>
    <li class="page-item active"><a class="page-link" href="#">1</a></li>
    <li class="page-item"><a class="page-link" href="#">2</a></li>
```

```
    <li class="page-item"><a class="page-link" href="#">Next</a></li>
  </ul>
</nav>
```

- `.active` → current page
- `.disabled` → non-clickable item

3️⃣ Modals

Pop-up windows for forms, messages, or media.

```
<!-- Button trigger -->
<button class="btn btn-primary" data-bs-toggle="modal"
data-bs-target="#infoModal">
  Open Info
</button>


<!-- Modal -->
<div class="modal fade" id="infoModal" tabindex="-1">
  <div class="modal-dialog">
    <div class="modal-content">
      <div class="modal-header">
        <h5 class="modal-title">Booking Info</h5>
        <button type="button" class="btn-close"
data-bs-dismiss="modal"></button>
```

```
      </div>
      <div class="modal-body">
        Please fill in your details for the trip.
      </div>
      <div class="modal-footer">
        <button class="btn btn-secondary"
data-bs-dismiss="modal">Close</button>
        <button class="btn btn-primary">Save</button>
      </div>
    </div>
  </div>
</div>
```

- `data-bs-toggle="modal"` → Opens modal
- `.modal-dialog` → Container size

4️⃣ Accordion & Collapse

Accordion

```
<div class="accordion" id="faqAccordion">
  <div class="accordion-item">
    <h2 class="accordion-header">
```

```
      <button class="accordion-button" data-bs-toggle="collapse"
data-bs-target="#collapseOne">

        What is included in the tour?

      </button>

    </h2>

    <div id="collapseOne" class="accordion-collapse collapse show"
data-bs-parent="#faqAccordion">

      <div class="accordion-body">

        All meals, accommodations, and activities are included.

      </div>

    </div>

  </div>

</div>
```

- Only **one section open** at a time if `data-bs-parent` is used.

Simple Collapse

```
<button class="btn btn-primary" data-bs-toggle="collapse"
data-bs-target="#details">Show Details</button>

<div class="collapse" id="details">

  <div class="card card-body">

    Extra tour details go here.

  </div>

</div>
```

5 Carousel

Rotating slideshow for images.

```
<div id="travelCarousel" class="carousel slide" data-bs-ride="carousel">


  <div class="carousel-inner">

    <div class="carousel-item active">

      <img src="img1.jpg" class="d-block w-100" alt="Beach">

      <div class="carousel-caption d-none d-md-block">

        <h5>Tropical Beach</h5>

      </div>

    </div>

    <div class="carousel-item">

      <img src="img2.jpg" class="d-block w-100" alt="Mountains">

      <div class="carousel-caption d-none d-md-block">

        <h5>Mountain Adventure</h5>

      </div>

    </div>

  </div>


  <!-- Controls -->
```

```
  <button class="carousel-control-prev" data-bs-target="#travelCarousel"
data-bs-slide="prev">

    <span class="carousel-control-prev-icon"></span>

  </button>

  <button class="carousel-control-next" data-bs-target="#travelCarousel"
data-bs-slide="next">

    <span class="carousel-control-next-icon"></span>

  </button>

</div>
```

- `.carousel-item.active` → First slide shown by default
- `.carousel-caption` → Text over image

🧠 PART 8: Bootstrap Helpers & Utility Classes

These are **quick-use classes** that help you style and position elements without writing custom CSS.

1️⃣ Border Utilities

You can quickly add, remove, or style borders.

Class	Effect
`.border`	Adds a border on all sides
`.border-top`	Border only at top
`.border-0`	Removes border
`.border-primary`	Border with primary color
`.rounded`	Small rounded corners
`.rounded-circle`	Perfect circle (for images)

```
<img src="avatar.jpg" class="border border-3 border-primary rounded-circle" width="100">
```

Here:

- `border-3` → Border thickness
- `border-primary` → Bootstrap theme color border
- `rounded-circle` → Makes it round

2️⃣ Shadows

Bootstrap provides **box shadows**:

Class	Effect
`.shadow-none`	No shadow
`.shadow-sm`	Small shadow
`.shadow`	Regular shadow
`.shadow-lg`	Large shadow

```
<div class="p-3 shadow-lg bg-white">Large shadow box</div>
```

3️⃣ Position Utilities

Used for **absolute/relative/fixed/sticky** positioning.

Class	Effect
`.position-static`	Default position
`.position-relative`	Relative to parent
`.position-absolute`	Absolute inside parent
`.position-fixed`	Fixed to viewport
`.position-sticky`	Sticks until parent ends

Example — **Sticky Navbar**:

```
<nav class="navbar navbar-dark bg-dark position-sticky top-0 shadow">
  <div class="container">
    <a class="navbar-brand" href="#">StickyNav</a>
  </div>
</nav>
```

- `position-sticky top-0` → Stays at top during scroll
- `shadow` → Adds shadow below

4️⃣ Flex Utilities

Bootstrap's **Flexbox helpers** let you align and distribute items quickly.

Common Classes:

Class	Effect
`.d-flex`	Enables flexbox
`.flex-row` / `.flex-column`	Direction
`.justify-content-start / center / end / between / around`	Horizontal alignment
`.align-items-start / center / end`	Vertical alignment

Example:

```
<div class="d-flex justify-content-between align-items-center border p-3">
  <div>Logo</div>
  <div>Menu</div>
</div>
```

5️⃣ Display & Visibility Classes

Display Classes control how an element is shown:

- `.d-none` → Hide element
- `.d-block` → Show as block
- `.d-inline` → Show inline
- `.d-lg-none` → Hide on large+ screens
- `.d-none d-md-block` → Hide on mobile, show on tablet+

Visibility Classes:

- `.visible` → Force visible
- `.invisible` → Hide but keep space

🎥 Demo — Sticky Navbar with Shadow

```
<nav class="navbar navbar-expand-lg navbar-dark bg-dark position-sticky top-0 shadow-lg">

  <div class="container">

    <a class="navbar-brand" href="#">TravelSite</a>

    <button class="navbar-toggler" type="button" data-bs-toggle="collapse" data-bs-target="#navMenu">

      <span class="navbar-toggler-icon"></span>

    </button>
```

```
    <div class="collapse navbar-collapse" id="navMenu">

      <ul class="navbar-nav ms-auto">

        <li class="nav-item"><a class="nav-link" href="#">Home</a></li>

        <li class="nav-item"><a class="nav-link"
href="#">Destinations</a></li>

        <li class="nav-item"><a class="nav-link"
href="#">Contact</a></li>

      </ul>

    </div>

  </div>

</nav>



<div class="container mt-5">

  <p>Scroll down to see the sticky effect in action...</p>

  <div style="height:2000px"></div>

</div>
```

- Navbar **sticks** to the top when scrolling
- `.shadow-lg` makes it visually distinct